

PROJECT PHASE II - REVIEW 1

AI-POWERED HEALTH REMINDER SYSTEM AND EMERGENCY ALERT APP

TEAM MEMBERS

FEBA M AJI

SHIBANA NAZEER

SREELAKSHMI SAI

JYOTHYLAKSHMI BIJU

GUIDE

Mrs.PRETTY SEBASTIAN

ASSISTANT PROFESSOR

PROBLEM STATEMENT

- **Health Monitoring Challenges:**

Elderly and chronically ill individuals require constant assistance.

- **Issues With Traditional Reminder Systems:**

Provide static reminders without intelligent tracking or real-time monitoring.

- **Risk of Undetected Emergencies :**

Missed medication and prolonged inactivity often go unnoticed.

- **Lack of User Friendly Features :**

Most apps are not designed with elderly accessibility.

MOTIVATION

- Elderly and sick people may forget to take medicines on time.
- Missing medicines or help during emergencies can be dangerous.
- Normal reminder apps are not smart and do not detect emergencies.
- Many apps are hard for elderly people to use.
- There is a need for a simple and smart health app.
- Using AI and sensors can help detect emergencies quick.

OBJECTIVES

- Enhance independence and safety for elderly and chronically ill individuals.
- Provide a user-friendly mobile app with intelligent medicine reminders.
- Enable real-time emergency detection using sensors (accelerometer).
- Facilitate caregiver support through a live monitoring dashboard.
- Incorporate voice commands for hands-free navigation.
- Use facial recognition for alert cancellation to ensure safety.
- Automate appointment reminders and manage healthcare provider visits.
- Ensure timely medication adherence and track missed doses.
- Monitor medicine stock .

KEY PAPERS

1. Intelligent and Safe Medication Box In Health IoT Platform for Medication Monitoring System with Timely Reminders.
2. MEMO Box: Health Assistant for Depression with Medicine Carrier and Exercise Adjustment Driven by Edge AI .
3. IoT Based Medical Assistant for Efficient Monitoring of Patients in Response to COVID-19.
4. ACTIVA: Innovation in Quality of Care for Nursing Homes through Activity Recognition.
5. Voice Assistants for Health Self-Management: Designing for and with Older Adults.

RESEARCH GAP

1. Intelligent and Safe Medication Box In Health IoT Platform for Medication Monitoring System with Timely Reminders.

- Lacks emergency detection (inactivity).
- No voice/multilingual support.
- No facial recognition Limited caregiver dashboard.
- No doctor/appointment reminders.
- Incomplete stock tracking.
- Limited elderly accessibility.

2. MEMO Box: Health Assistant for Depression with Medicine Carrier and Exercise Adjustment Driven by Edge AI .

- No caregiver dashboard, voice navigation.
- Lacks real-time emergency detection (falls, SOS).
- Focuses on depression management, not general elderly care.

3. IoT Based Medical Assistant for Efficient Monitoring of Patients in Response to COVID-19.

- Hardware-heavy (sensors + microcontrollers).
- No caregiver dashboard or mobile family alerts.
- No accessibility features (multilingual, SOS shake/face cancel).
- Limited to COVID, not broader elderly/chronic carebody text.

4. ACTIVA: Innovation in Quality of Care for Nursing Homes through Activity Recognition.

- No medicine reminders.
- No AI predictions (just rules).
- No voice or multilingual support.
- No face-based alert cancellation.
- Caregiver focus only, not family.

5. Voice Assistants for Health Self-Management: Designing for and with Older Adults.

- No emergency alert system.
- No stock tracking or doctor scheduling.
- No face-based alert cancellation.
- Only for elderly, not all ages.
- No family caregiver dashboard.

SUMMARY TABLE

<i>REF</i>	<i>TITLE</i>	<i>KEY FEATURES</i>	<i>GAPS IDENTIFIED</i>
[1]	Intelligent and Safe Medication Box In Health IoT Platform for Medication Monitoring System with Timely Reminders - IEEE, 2021	• Smart medication box with 6 sub-boxes for organizing pills. • Timely reminders via Android app for time, dosage, and stock. • Bio-sensors for temperature and heartbeat monitoring. • Authentication to prevent misuse/overdosage.	• Lacks emergency detection (falls, inactivity). • No voice/multilingual support. • No facial recognition • Limited caregiver dashboard. • No doctor/appointment reminders. • Incomplete stock tracking. • Limited elderly accessibility.
[2]	MEMO Box: Health Assistant for Depression with Medicine Carrier and Exercise Adjustment Driven by Edge AI – IEEE, 2023	• Smart medicine box with medication management. • Emotion recognition using AI & EEG data. • Exercise recommendation through reinforcement learning. • Edge & cloud integration for low latency response.	• Focuses on depression management, not general elderly care. • Requires dedicated hardware (box, wearables). • No caregiver dashboard, multilingual voice navigation. • Lacks real-time emergency detection (falls, SOS).

<i>REF</i>	<i>TITLE</i>	<i>KEY FEATURES</i>	<i>GAPS IDENTIFIED</i>
[3]	IoT Based Medical Assistant for Efficient Monitoring of Patients in Response to COVID-19 – IEEE, 2020	• Biomedical sensors: heartbeat, temperature, motion, saline level. • Data auto-uploaded to cloud (OneDrive) every 30s. • Automated alarms for abnormal readings. • Android app for bed control, reminders, nurse call. • Enables remote monitoring, reduces infection risk.	• Designed mainly for hospital COVID-19 monitoring. • Hardware-heavy (sensors + microcontrollers). • No caregiver dashboard or mobile family alerts. • No accessibility features (multilingual, SOS shake/face cancel). • Limited to COVID, not broader elderly/chronic care.
[4]	S. González-Mendoza et al., “ACTIVA: Innovation in Quality of Care for Nursing Homes through Activity Recognition,” IEEE Access 2023.	• Uses HAR (activity recognition) for elderly. • Sensors to detect activities. • Sends real-time alerts to caregivers.	• Works only in nursing homes. • No medicine reminders. • No AI predictions (just rules). • No voice or multilingual support. • No face-based alert cancellation. • Caregiver focus only, not family.

<i>REF</i>	<i>TITLE</i>	<i>KEY FEATURES</i>	<i>GAPS IDENTIFIED</i>
[5]	M. Sezgin et al., "Voice Assistants for Health Self-Management: Designing for and with Older Adults," Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems, Hamburg, Germany, Apr. 2023.	• Voice assistant for elderly care. • Medicine reminders & health info. • Easy language interaction.	• Only voice, no multimodal (face/UI). • No emergency alert system. • No stock tracking or doctor scheduling. • No face-based alert cancellation. • Only for elderly, not all ages. • No family caregiver dashboard.
[6]	H. Tian et al., "Health Monitoring and Automatic Notification Device Based on Laser-Induced Graphene", IEEE Sensors Journal, Jul. 2019.	• LIG sensors for respiration monitoring. • Detects apnea & abnormal breathing. • Sends automatic health alerts.	• Only respiration monitoring • No medicine reminders • No voice or multilingual support • No face detection alert cancel • No doctor/medicine management • Hardware-only focus

<i>REF</i>	<i>TITLE</i>	<i>KEY FEATURES</i>	<i>GAPS IDENTIFIED</i>
[7]	M. Alberdi, A. Aztiria, and A. Basarab, “Designing Interactive Alerts to Improve Recognition of Critical Events in Medical Emergencies,” Proceedings of the 2019 ACM Conference on Human Factors in Computing Systems (CHI), Glasgow, UK, May 2019.	• Designs interactive visual alerts for medical staff. • Improves recognition of critical events in trauma/emergency situations. • Helps reduce missed vital sign documentation errors.	• Focus only on hospital staff alerts. • Provides visual alerts only. • No medicine reminders. • No emergency SOS triggers (shake/inactivity). • No caregiver/family dashboard. • Not AI-driven.
[8]	Y. Baek and L. Gualtieri, “Reimagining Reminders to Increase Medication Adherence in Older Adults,” ACM MobiCom4AgeTech ’24, Washington D.C., USA.	• Interviews with 22 elderly (50–87 yrs). • Most used routine-based triggers (coffee, bedtime). • Found notification fatigue from time-based alerts. • Suggested as-needed/routine reminders.	• No real system. • No IoT sensors/emergency alerts. • No caregiver dashboard. • Only reminders. • No voice/facial support.

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<i>REF</i>	<i>TITLE</i>	<i>KEY FEATURES</i>	<i>GAPS IDENTIFIED</i>
[9]	R. Rajat, C. Manohar, H. Paliwal, N. Bedekar, R. Bajoria, and V. Aggarwal, "Automated Educative Nudges to Reduce Missed Assignments in College," 2021.	• AI-based system to send reminders (nudges). • Helped students complete tasks on time. • Focus on behavior change through reminders.	• Education focus. • No IoT sensors or emergency alerts. • No caregiver dashboard. • Only text nudges. • No medicine tracking.
[10]	Agarwal, V. Sakthivel, and P. Prakash, "Toward Inclusive Healthcare: An LLM-Based Multimodal Chatbot for Preliminary Diagnosis," 2025.	• Wearable device with graphene sensors. • Tracks pulse, movement, body signals. • Sends automatic health alerts. • Low-cost and flexible design.	• Hardware only. • No medicine reminders. • No multilingual support. • No caregiver dashboard. • No SOS/facial recognition.

<i>REF</i>	<i>TITLE</i>	<i>KEY FEATURES</i>	<i>GAPS IDENTIFIED</i>
[11]	Qiu et al., "Large AI Models in Health Informatics: Applications, Challenges, and the Future," IEEE Journal of Biomedical and Health Informatics, vol. 27, no. 12, Dec. 2023	• Large-scale AI models • Multi-modal data handling • Zero-/few-shot learning • Generalist intelligence • Generative abilities • 7 domains: bioinformatics, diagnosis, imaging, informatics, education, public health, robotics	• Large-scale research. • Hospital/research. • High computational cost. • Needs massive annotated. • Generalized AI. • Future-oriented prototypes.
[12]	Z. Dong, J. Wei, X. Chen, and P. Zheng, "Face Detection in Security Monitoring Based on Artificial Intelligence Video Retrieval Technology," IEEE Access, April 2020.	• AI & video retrieval for face detection. • Real-time and accurate. • Works with low-light faces. • Uses deep learning. • Combines global and local features. • For security monitoring.	• Only for security, not for healthcare. • No medicine reminders. • No emergency alerts. • No caregiver dashboard. • No voice or elderly-friendly features.

PROPOSED SOLUTION

Methodology

1. User Data Collection

- Collects essential details such as medicine information, doctor appointments, and caregiver contacts.
- Captures user interactions including button presses, voice commands, and SOS actions through the app.

2.Doctor Scheduling & Appointment Reminder Module

- Allows users to schedule and manage doctor appointments.
- Sends reminder notifications and maintains appointment history.

3.Medicine Reminder Module

- Sends timely alarms and notifications for medicine intake.
- Records missed or delayed doses for caregiver monitoring.

4. Emergency Alert Module

- Enables users to trigger SOS manually using a button or gesture.
- Immediately sends emergency alerts to the backend system.

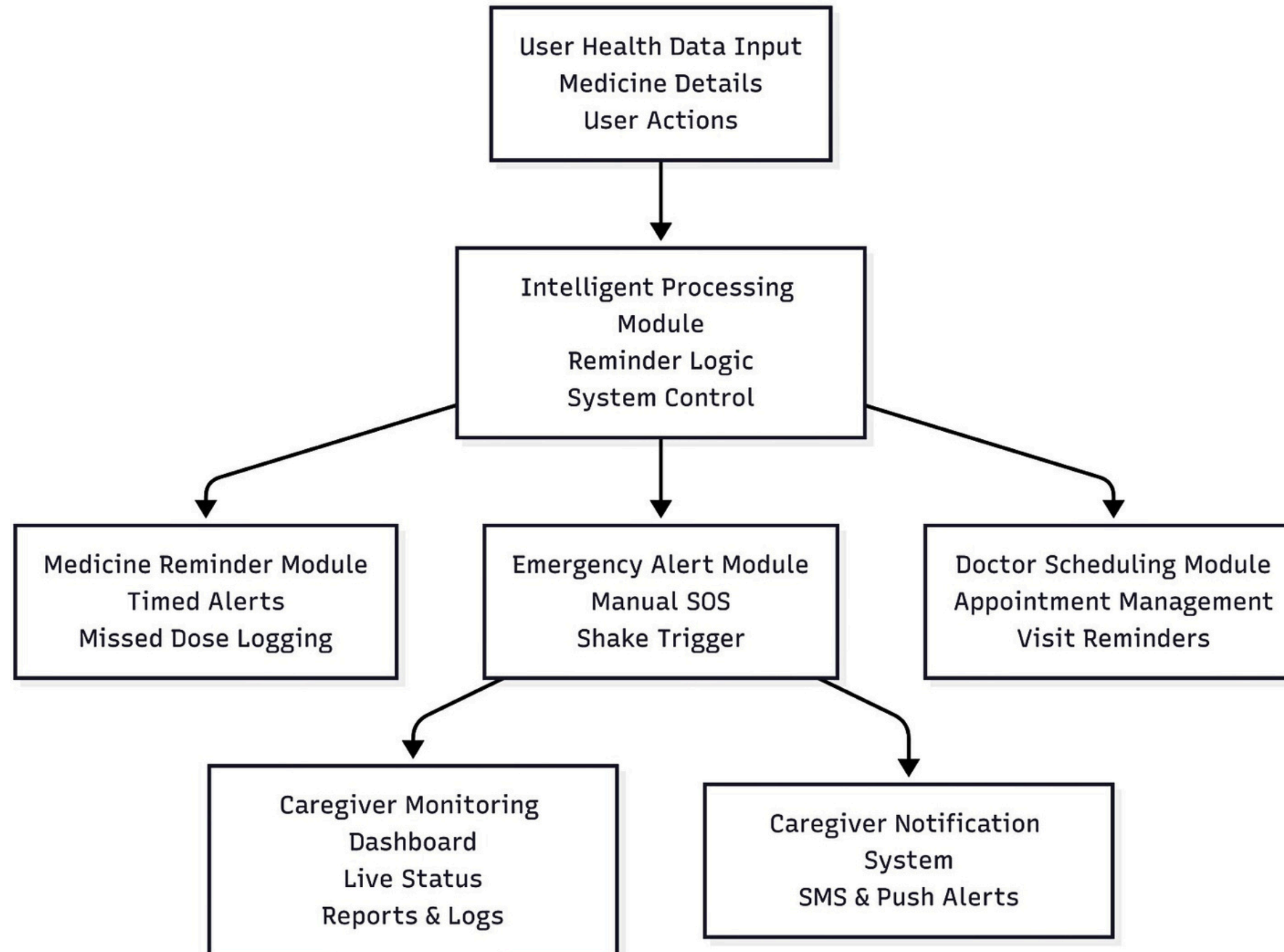
5. Caregiver Notification & Monitoring

- Sends SMS and push notifications during emergencies or missed medicines.
- Allows caregivers to monitor user status through reports and dashboards.

6. Voice Navigation Module

- Supports voice commands.
- Enables elderly-friendly, hands-free navigation.
- CNN language is used for voice emotion.

PROPOSED SOLUTION



NOVELTY & SCOPE

Innovation

- AI-based medicine reminder and emergency detection system.
- Uses smartphone sensors for inactivity detection.
- Face recognition to cancel false emergency alerts.
- Voice-enabled elderly-friendly interface.

Implementation Scope

- Mobile app for elderly and chronically ill users.
- AI logic for reminders and emergency alerts.
- Cloud backend for data storage and notifications.
- Caregiver dashboard for remote monitoring.

Advantages

- Improves user safety and independence.
- Reduces missed medicines and false alerts.
- Easy to use with voice support.
- Cost-effective using existing smartphone hardware.

TASK DISTRIBUTION

Team member roles

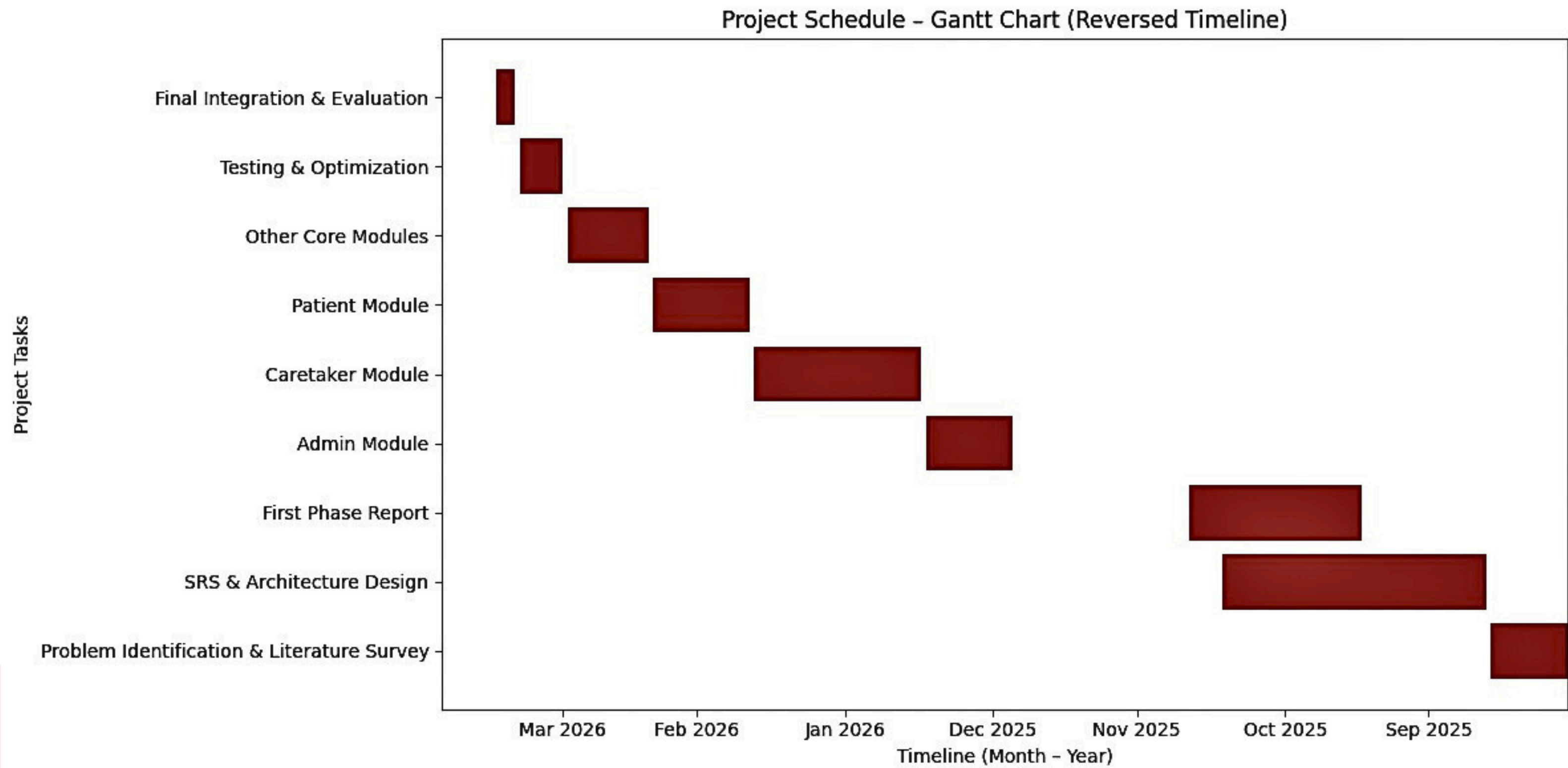
1. Sreelakshmi Sai - Login , Dashboard & Emergency Detection
2. Jyothylakshmi Biju - Patient , Caregiver Data & Voice Navigation
3. Feba M Aji - Mobile Application , AI Chatbot & Doctor Appointment
4. Shibana Nazeer - Testing , Documentation & Face Detection

Work Breakdown

- Login , Dashboard & Emergency Detection
 - > Created login page
 - > User roles (Caregiver, Family, Medical staff)
 - > Dashboard screen design
 - > Alert system and cancellation

- Patient , Caregiver Data & Voice Navigation
 - > Added patient details
 - > Caregiver management
 - > View, add, and delete patients
 - > Used for spoken commands
- Mobile Application , AI Chatbot & Doctor Appointment
 - > Patient registration in mobile & appointments of the doctor
 - > Manage patient's screen
 - > Medicine button and alerts
 - > Manage and simplify daily activities
- Testing , Documentation & Face Detection
 - > Tested all features
 - > Fixed errors
 - > Find emotions and continues monitoring

PROJECT SCHEDULE



Milestones

- > Problem identification and literature survey - Completed.
- > SRS and architecture design - Completed.
- > First phase report - Completed
- > Admin Module - Completed
- > Caretaker Module - Completed
- > Patient Module - Completed
- > Other core module implementations - Completed
- > Testing and optimization - Upcoming
- > Final integration and evaluation - Planned

INTERIM RESULTS

Current Progress

- Admin module implemented
- Mobile app UI development implemented.
- Medicine reminder and scheduling module implemented.
- Sensor-based inactivity detection implemented.
- Voice commands implemented.
- Backend and caregiver dashboard implemented.
- Emergency alert integration implimented.

Screenshots



Welcome to CareTrack

Sign in to access the patient monitoring dashboard


Caregiver


Family Member


Medical Staff

Email Address

 admin@gmail.com

Password

 Enter your password 

☐ Remember me [Forgot password?](#)

Sign In

OR

Real-time Health Monitoring

Track vital signs, medication schedules, and activity patterns with our innovative system designed specifically for dementia care.

Monitor Health Data



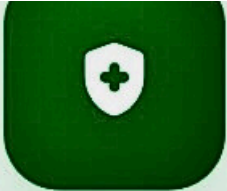
Patient Management

 **Rajesh D**
Professional Caregiver - FamilyMember object (3)

ID:	6
Phone:	7896852356
Email:	rajesh@gmail.coml
Date of Birth:	22/3/1973
Gender:	Male

System Features

-  Health Monitoring
-  Location Tracking



Welcome to MindCare Connect

Dementia Care Management System

Healthcare Login

Enter your credentials to access dementia care management

SIGN IN

New to MindCare?

CAREGIVER

Caregiver Registration

Register as a caregiver to provide support and manage dementia care

Gender *

Male

Address *

Enter your complete address

Gender *

Male

Address *

Phone Number *

Gender *

Male

Address *



MindCare Connect

Dementia Care Management System

Connect to your healthcare server to access patient care tools



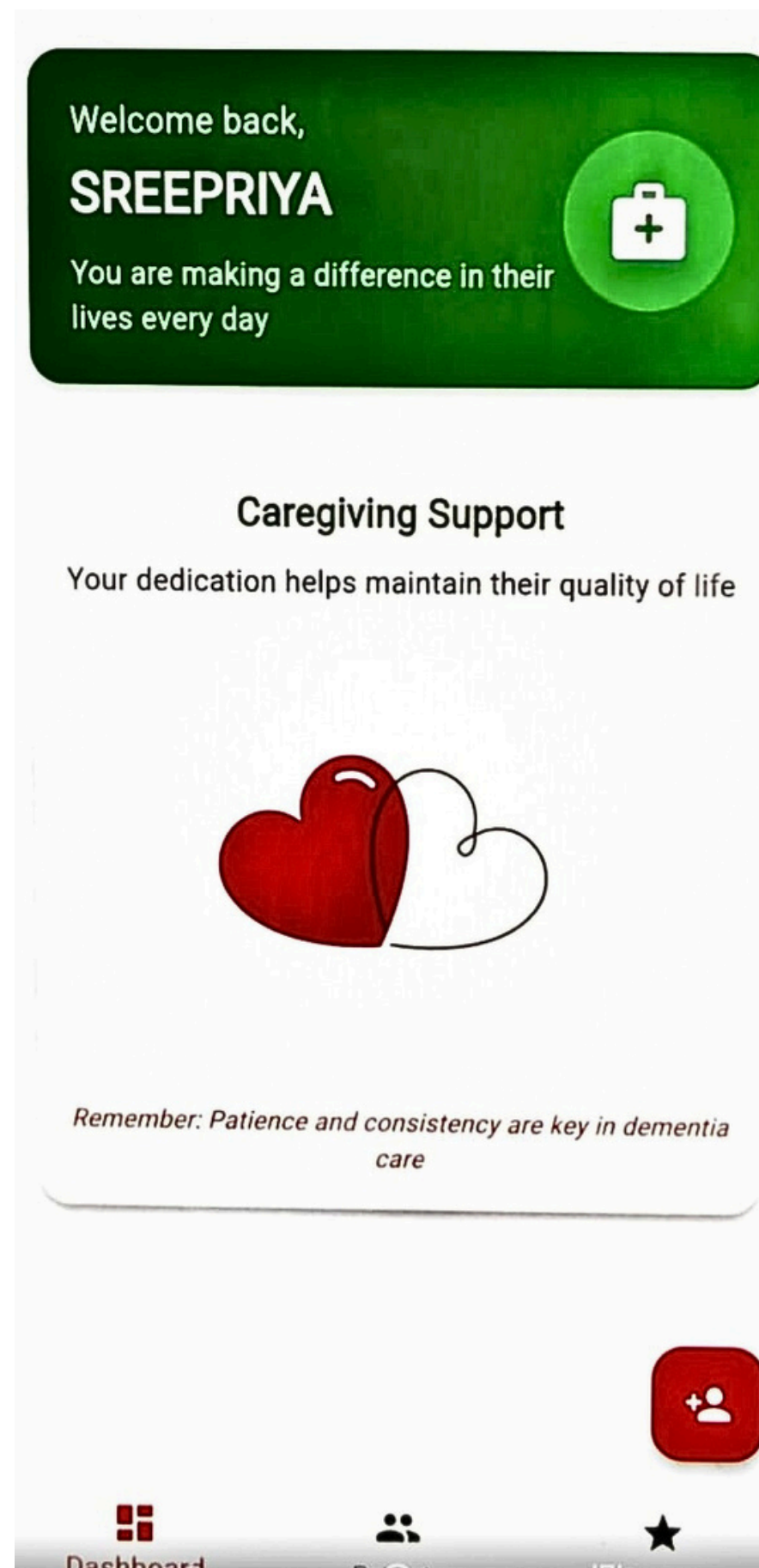
Connected Successfully!

Ready to access dementia care management tools

ACCESS PATIENT CARE

CONNECT TO HEALTHCARE

 Need help finding your server IP?



← Patient Registration

Add Profile Photo

Dementia Care Patient Registration

Create your healthcare account to access personalized care

Full Name *

Email Address *

Phone Number *

Date of Birth *

Gender *

Male

Relationship to Primary Caregiver *

← Manage Patients

Rajesh D

7896852356

ID: 6

Date of Birth: 22/3/1973

Gender: Male

Email: rajesh@gmail.com

Relationship: Professional Caregiver

Sleep Hours: 22:00:00 - 06:00:00

Address:

Sreelakam

House No. 15

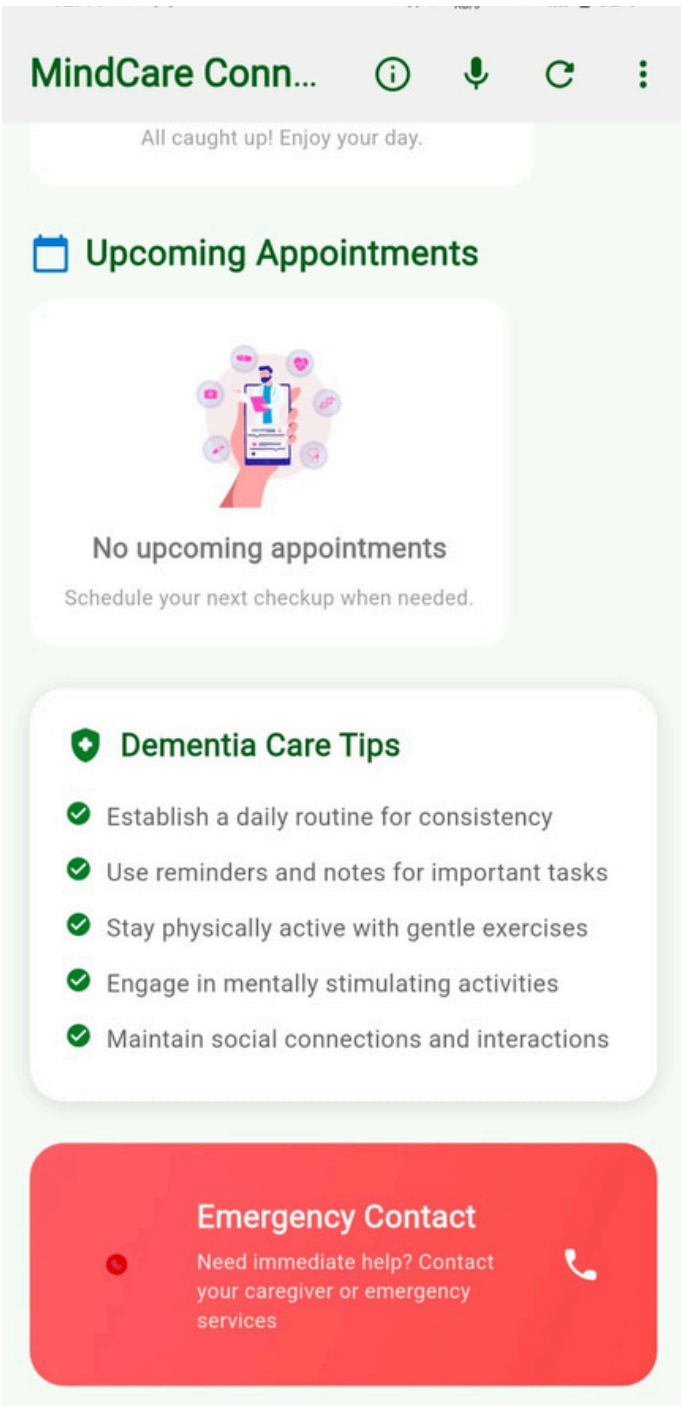
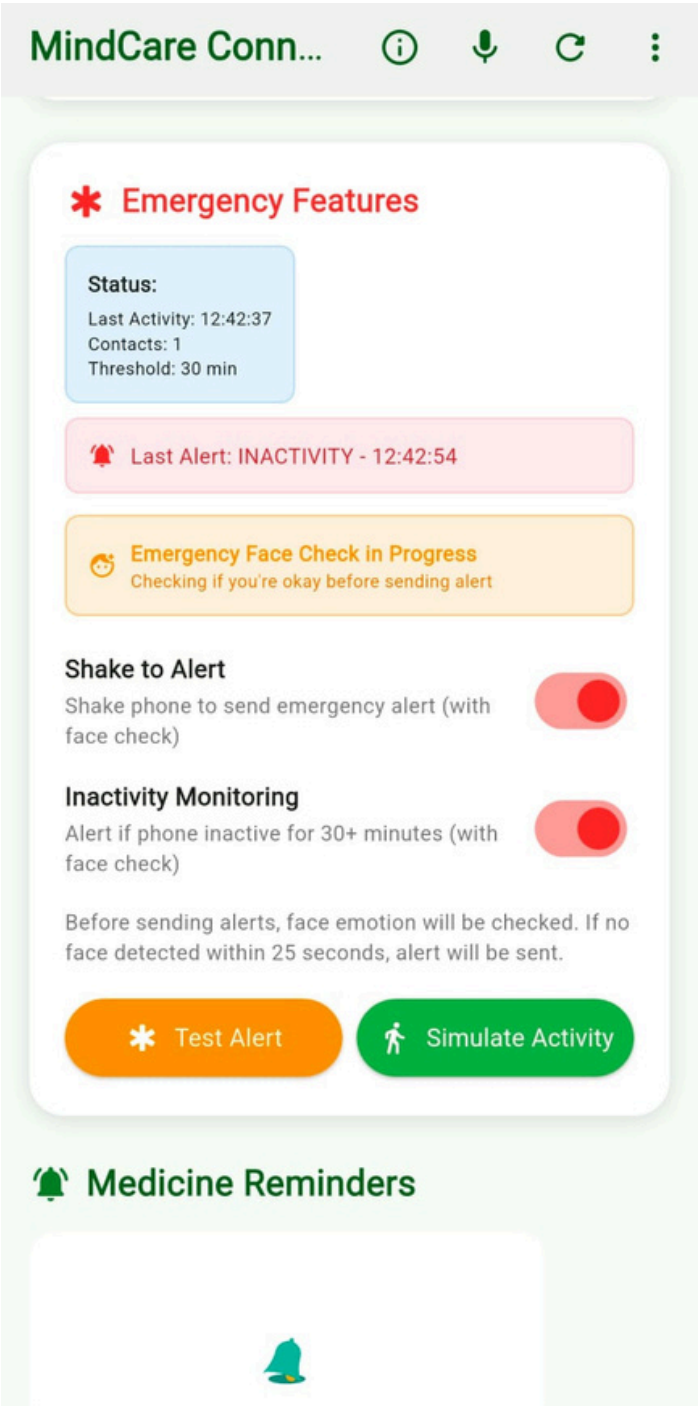
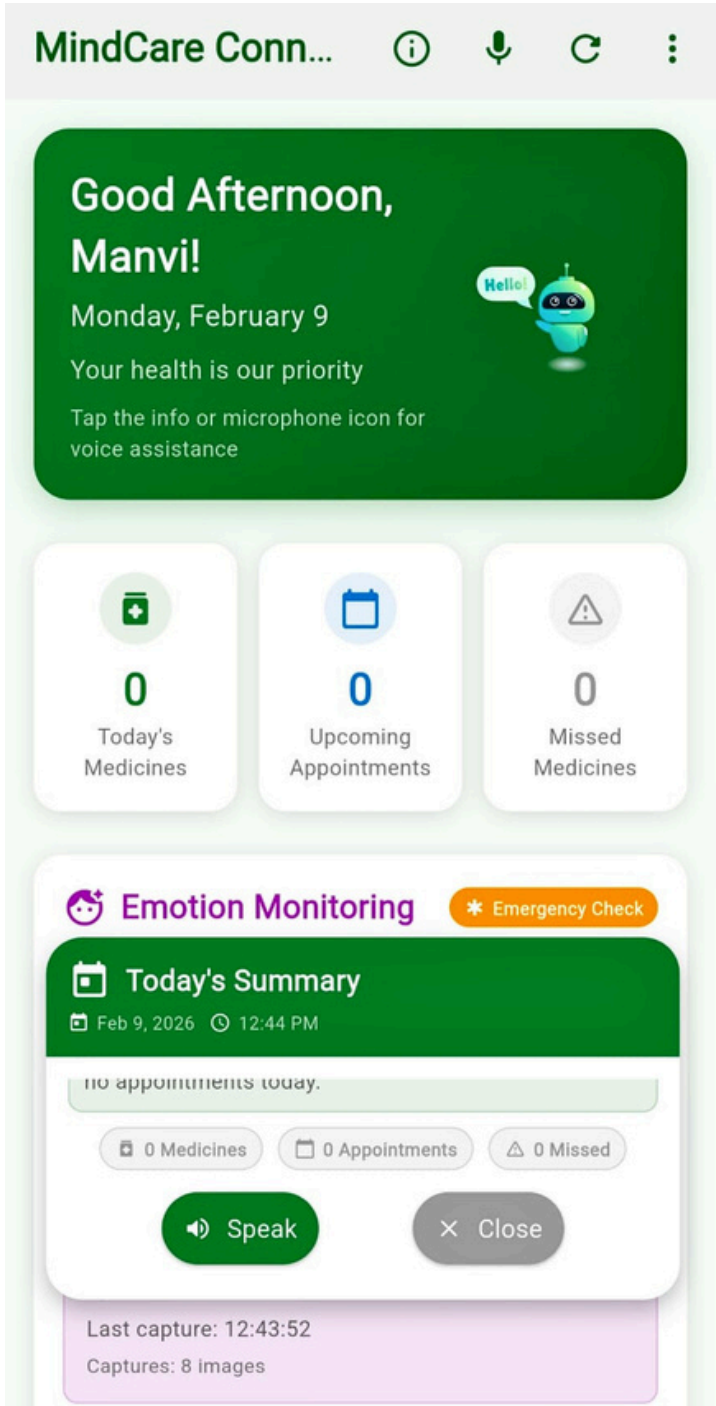
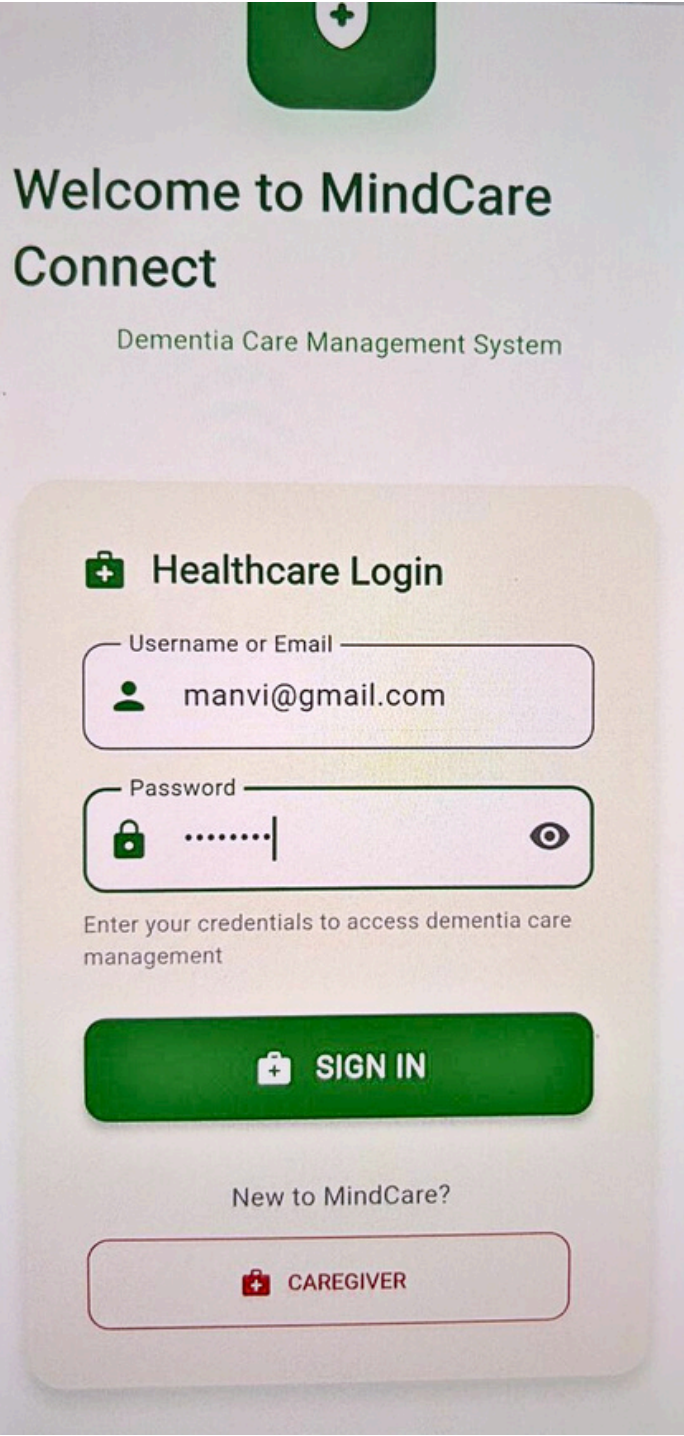
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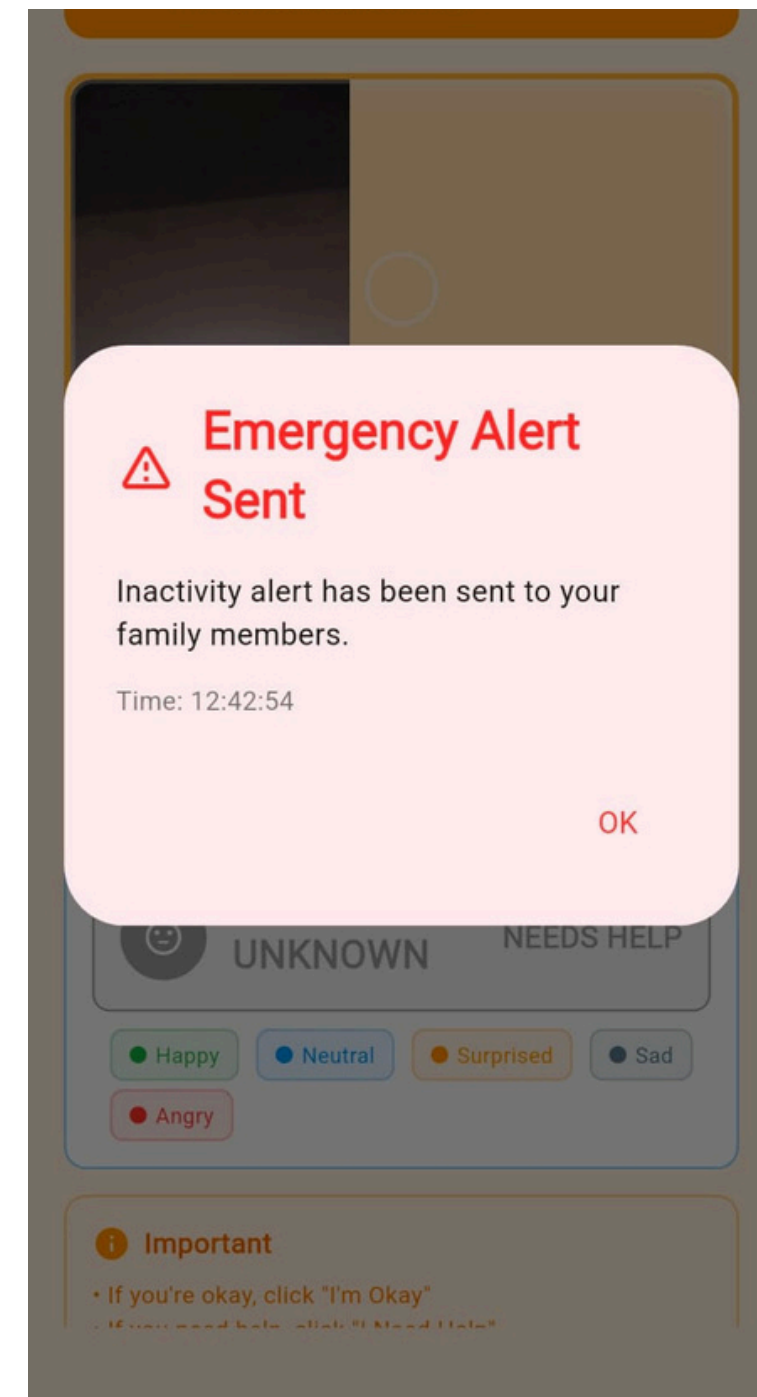
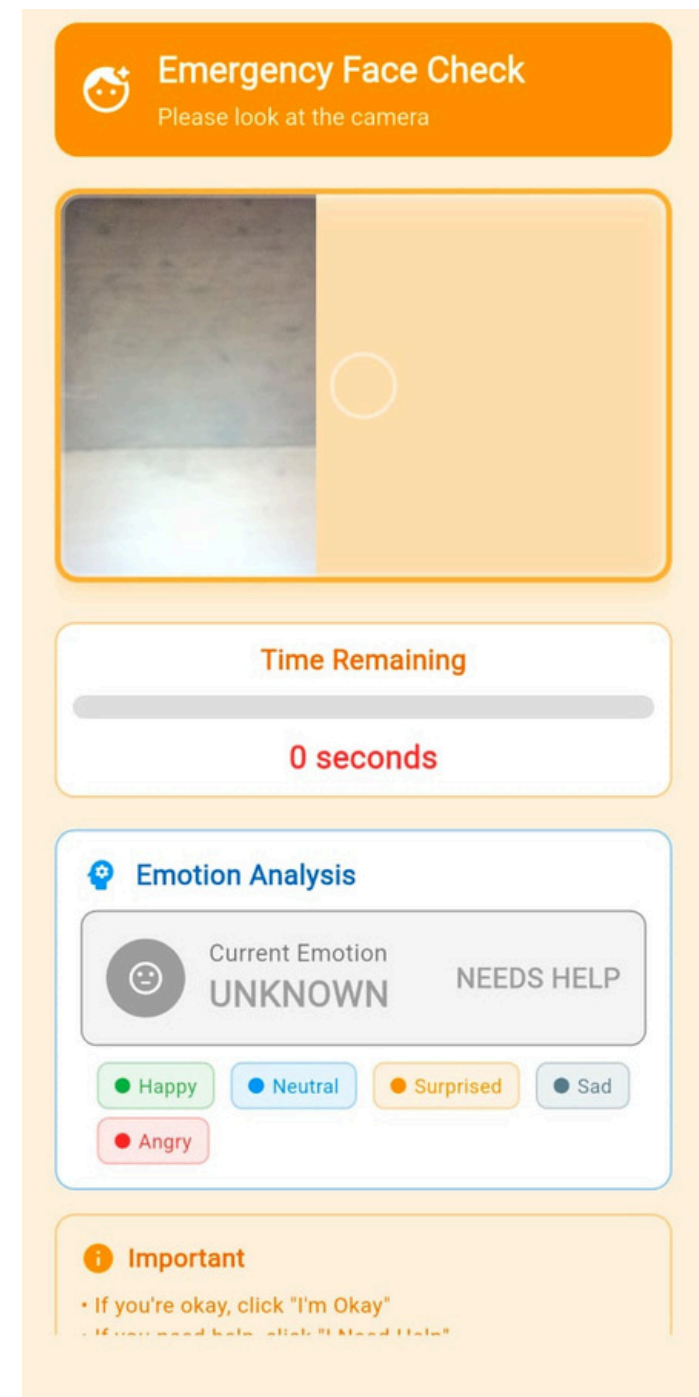
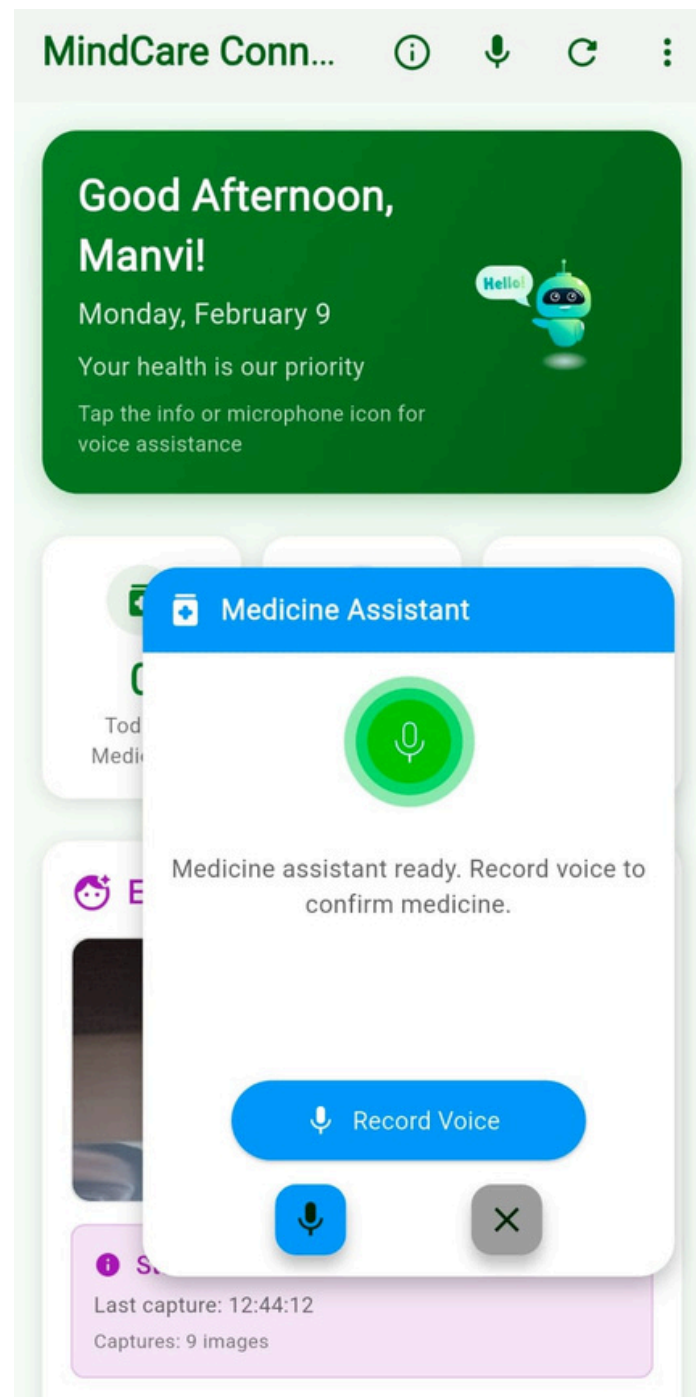
Kerala

India

Medicine

Delete





CHALLENGES & RISK

Technical Challenges

- Detecting Emergencies Correctly
- Sensor Accuracy
- App Performance
- Internet Dependency
- Data Storage and Security

Mitigation Plan

- Check movement more than once before sending alert.
- Ignore small unwanted movements.
- Reduce background work to keep app smooth.
- Send SMS alerts when internet is not available.
- Protect data using login and secure storage.

CONCLUSION

Summary.

- AI-powered system for health reminders and emergency alerts
- Improves medication adherence and user safety
- Enables real-time caregiver notifications
- User-friendly design focused on elderly needs
- Scalable for future healthcare enhancements

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THANK YOU